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Research Paper

Knowledge enhanced multi-task learning for simultaneous optimization of human parsing and pose estimation

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ABSTRACT

Human parsing and pose estimation are both closely related to the human body structure, and it is advantageous to deal with both together in order to assist each other's learning. Nevertheless, the two tasks have their own unique characteristics, and it is therefore difficult to efficiently and simultaneously obtain a good performance for both tasks in one single multi-task learning (MTL) network. This paper proposes a new MTL network (named "KDNet") to enhance the communication between human parsing and pose estimation through knowledge distillation so as to improve the simultaneous learning of both tasks. In the proposed KDNet, a consistent representation module is used to enhance the related information sharing between the tasks, and a single-task model-based loss weighting method is developed to balance the loss levels. The proposed KDNet has been verified across three benchmark datasets, and in comparison to other state-of-the-art methods it achieves substantial performance gains without increasing model parameters during inference. The codes are shared at https://github.com/Yhdian/KDNet.git.

1. Introduction

Human parsing and pose estimation represent very challenging problems and have been extensively studied in recent years, due to their great value in human-centred applications, such as human motion analysis, human–computer interaction, biomechanics and medication, virtual try-on of fashion, etc. For instance, for virtual try-on application, simultaneous estimation of human parsing and human pose is crucial for realistic and accurate virtual try-on, ensuring the virtual garments align properly with the body's movements and posture. Human parsing generates classification score maps and assigns a class label to each pixel of the input image. Pose estimation generates joint probability maps in the form of heatmaps and then locates pixel locations of joint keypoints on the heatmaps. Both of these tasks obtain pixel-level score/probability maps and learn the representation of the human body structure, so that they can be leveraged to assist each other in the process.

In previous network-related studies of this nature, the features for the two tasks are learned individually and their relation is then modelled to improve each other. For example, being the first study integrating pose estimation with semantic part segmentation, Xia et al. (2017) first trained two separate fully convolutional neural networks and then updated the networks with initial predictions as prior and constraints. In order to combine the two tasks in an end-to-end framework, Liang et al. (2018) shared a CNN for feature learning of both tasks, used separated modules for their prediction and a refinement network to produce more accurate results using generated context and prediction maps. Nie et al. (2018a) used separate CNNs to learn features for human parsing and pose estimation, and the learned features were used by a mutual adaptation module to guide each other. Considering the spatial relationship between the two tasks, Zhang et al. (2020c) modelled such a spatial relationship using a self-attention scheme (Wang et al., 2018). None of these studies, however, were intended to achieve good performance and efficiency for both tasks simultaneously. To perform both tasks in one single model, Zeng et al. (2021) designed separated fusion branches to combine features from specific tasks. Additionally, Neural Architecture Search (NAS) was utilized to search the best architecture for these fusion branches. Nevertheless, all these studies introduced additional module(s) for feature fusion in order to assist multi-task learning, improving the performance with compromised model efficiency or adaptability.

The difficulties of achieving ***both good and efficient*** performance on ***both tasks*** of human parsing and pose estimation in ***a single network***

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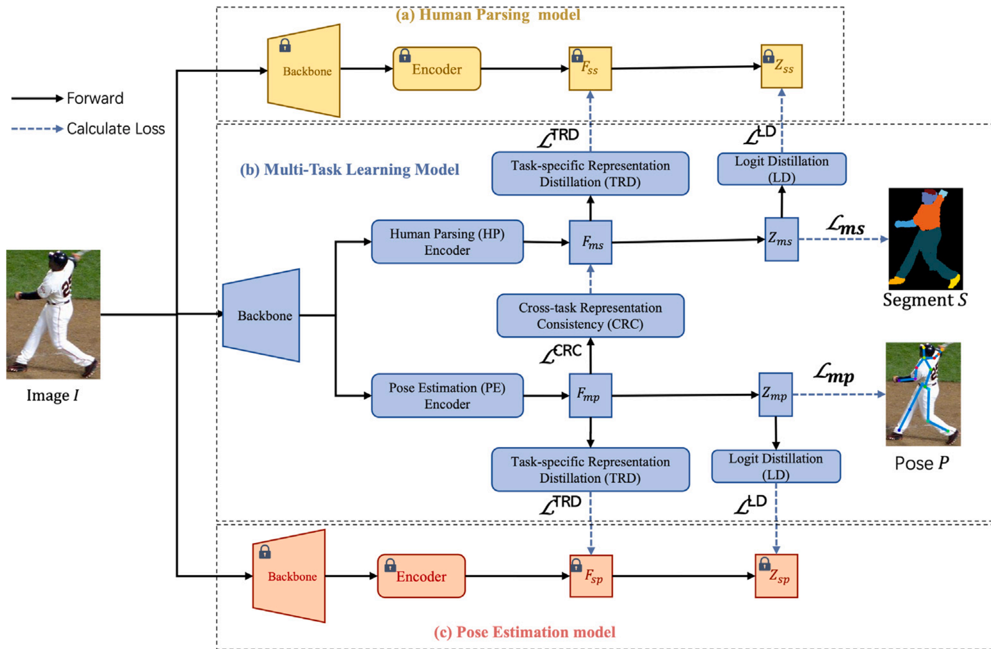

Fig. 1. The framework of "KDNet" — the proposed knowledge distillation based method for multi-task learning of human parsing and pose estimation. The proposed multi-task learning model, as shown in blue colour denoted as (b), consists of task-specific representation distillation (TRD) modules, logit distillation (LD) modules and a cross-task representation consistency (CRC) module. To train this model, two single task-specific models are first trained for human parsing, as shown in yellow colour denoted as (a), and for pose estimation, shown in red colour denoted as (c). Next, the multi-task learning model (b) is optimized using losses $\mathcal{L}_{ms}$, $\mathcal{L}_{mp}$, $\mathcal{L}^{TRD}$, $\mathcal{L}^{LD}$ and $\mathcal{L}^{CRC}$. During the inference phase, TRD and CRC modules are removed from the model.

are twofold. Firstly, it is difficult to efficiently leverage each other's information for mutual benefit. Existing work (Liang et al., 2018; Nie et al., 2018a; Zhang et al., 2020c; Zeng et al., 2021) design, manually or automatically, fusion or adaptation modules to interact the features of the two tasks. To do so, additional parameters are introduced to learn the relationship between human parsing and pose estimation in order to improve the targeted task. Secondly, it is hard to balance the training of two tasks since human parsing and pose estimation differ in their levels of difficulty. The traditional way is to search for optimal weights of their losses in a continuous hyper-parameter space, which is prohibitively expensive. To address the imbalance issue, recent studies proposed to adjust loss weights periodically during the training based on the uncertainty of the losses (Kendall et al., 2018), learning speed (Chen et al., 2018b; Liu et al., 2019a), task gradient (Chen et al., 2018b), key performance indicators (Guo et al., 2018), and loss scales (Lee et al., 2021). On the other hand, Li and Bilen (2020) proposed the alignment of the features from the single task models and MTL network, based on knowledge distillation (KD). Regardless of all these methods, it is still required to define initial loss weights, with equal weights are usually being used as default. Nevertheless, it can be argued that the setting of the initial loss weights is vital to balance the network training when the loss values of two tasks are greatly different.

This paper designs a novel MTL network based on KD technology, denoted as 'KDNet', to improve the learning of human parsing and pose estimation simultaneously, by strengthening the ***communication*** between the two tasks. Different from all other existing approaches introducing fusion module(s), this MTL network distills knowledge from the single-task network into not only the representation of the corresponding task but also, through a cross-task representation consistency (CRC) module, the representation of the other task. In this way, the knowledge of the two tasks can be effectively communicated, which contributes to best task-specific and shared representations. It is important to note that this CRC module is only required in the training phase and is removed in the inference phase, hence it will not increase the inference phase parameters or the computation. Additionally, KDNet, without altering the network structure, can be seamlessly integrated with any state-of-the-art approach for pose estimation and/or human parsing to enhance both simultaneously. Moreover, a loss weight algorithm based on the analysis of the loss variation of task-specific models is proposed to set the weights of the task losses, balancing the learning of two tasks. The main contributions of the present work are as follows:

- A multi-task network KDNet, based on knowledge distillation, is developed for simultaneous human parsing and pose estimation. Instead of optimizing a main task by learning an auxiliary task in most existing MTL studies, the proposed network is ***truly multi-tasking*** that optimizes both tasks in parallel, outperforming the best task-specific models both in terms of accuracy and efficiency.
- A cross-task representation consistency (CRC) module is created to efficiently guide the learning of human parsing and pose estimation, respectively, without increasing the number of parameters or computations involved in the inference phase.
- A task-specific model based loss weighting method is developed to automatically compute the loss weights for both tasks during the training process.
- The KDNet has been evaluated using three benchmark datasets, namely the LIP, the extended Pascal-Person-Part (ePPP), and the MPII datasets, and it achieves state-of-the-art performance on both the human parsing and pose estimation tasks. For human parsing, the present approach achieved 60.16% in terms of mean Intersection-Over-Union (IoU) on the LIP dataset and 72.93 on the ePPP dataset. For pose estimation, it achieved 88.6% in terms of PCK on the LIP dataset and 91.0% on the MPII validation dataset and 47.6% in terms of mAP on the ePPP dataset.
- The KDNet model is verified to have a relatively small model size and low computational complexity, compared to other state-of-the-art models, with further advantages of wide applicability and flexible structure, being effective for either parallel or one-way distillation. Although KDNet is specifically applied to the multi-task learning of human parsing and pose estimation, its underlying architecture and principles have the potential to be extended to other multi-task learning scenarios because of its flexible and adaptive design.

2. Related work

2.1. MTL network

Multi-task learning (MTL) is to learn multiple tasks simultaneously in a shared network so as to improve data efficiency and fast learning, and to reduce overfitting. For MTL, different tasks share some of the parameters through *hard parameter sharing*, or have their own models while sharing information by minimizing the distance between the model parameters of the different tasks, which is called *soft parameter sharing*. When using individual models for different tasks, soft parameter sharing enables the model to learn distinctive representations for different tasks and thereby performs better than hard parameter sharing, but it may result in overfitting due to the large number of parameters, which arise from multiple full models and will ultimately lead to high computational cost during the inference process. Typical methods of soft parameter sharing include Cross-stitch (Misra et al., 2016), Sluice (Ruder et al., 2019) and NDDR (Gao et al., 2019). For hard parameter sharing, all tasks share part of the model parameters, which is computationally less costly than soft parameter sharing under the same architecture.

Irrespective of hard or soft parameter sharing, MTL may encounter conflicts in terms of parameter updating because of the different objectives of the two tasks while the parameters are being shared. To solve this problem, some researchers suggested branching within the network to adjust where to share, some suggested task grouping without sharing any parameters, and other suggested balancing the training. Kendall et al. (2018) maximized the Gaussian likelihood with the task-dependent uncertainty, and the degree of uncertainty was used as the loss weight. Chen et al. (2018b) proposed GradNorm that opts for an equal learning speed of all tasks and adaptively computes loss weights by minimizing the distance between the desired task gradient and the average task gradient. GradNorm calculates the task gradient for every task using backpropagation, and it is computationally costly, even though it is only being applied to the last shared layer. DWA (Liu et al., 2019a), on the other hand, sets task weights based on task loss values without calculating the task gradient, as with GradNorm, all its tasks have a similar learning speed. Instead of balancing learning speed, Guo et al. (2018) used task difficulty to set loss weights in DTP, in which a key performance indicator (KPI) was used to quantify task difficulty, with higher weights being assigned to more difficult tasks. Since both DWA (Liu et al., 2019a) and DTP (Guo et al., 2018) are solely based on the learning speed, the learning may be dominated by tasks with large loss values. To balance the loss contribution, LSB (Lee et al., 2021) adjusted loss weights based on the loss scale, namely the product of the loss value and its weight, but assumed that the tasks have the same learning speed, which may not necessarily be true due to differences in the degree of difficulty and relevance of the different tasks. *In contrast to these previous approaches, this paper proposes that the loss weights be set based on the observed single tasks loss values and the loss variations. Within the MTL context, the proposed method optimizes two complementary tasks in parallel and simultaneously, rather than optimizing a main/target task with an auxiliary task in most MTL studies.*

2.2. Human parsing

Human parsing aims to segment from a human image the detailed regions of different clothing and the human body sections. To address the fine-grained segmentation task, some studies have modelled different levels of contexts on the basis of fully convolutional neural networks (FCNs). For example, Liang et al. (2015) designed a Co-CNN to capture the overall (global) and local super-pixel contexts present in the image. Ruan et al. (2019) used pyramid scene parsing (PSP) (Zhao et al., 2017a) to extract the multi-scale context and integrated edge information so as to improve the segmentation performance of the smaller items. Zhang et al. (2020b) proposed a Part-aware Context Network (PCNet) to extract the features of the human sections with various sizes as well as that of the clothing parts.

To take full advantage of the human body structure for parsing, some work have either modelled the human body structure or integrated with the pose information. Zhu et al. (2018) developed a progressive cognitive network to recognize different parts of the human body based on a component-aware region convolution structure. Wang et al. (2019) proposed to learn the relationship between body parts so as to refine the parsing results based on a hierarchical human body structure. Similarly, Li et al. (2020b) improved human parsing by means of a dual graph reasoning framework.

To integrate the human pose information, Nie et al. (2018a) proposed a mutual adaptation module to relate the pose and parsing learning, but the training process of their model was an iterative one. Liang et al. (2018) addressed the problems of pose estimation and human parsing using a multi-task learning approach, applying a refinement network to fuse the intermediate features and predictions by concatenation. Zhang et al. (2020c) investigated the relationship between parsing pixels and pose pixels, and used the calculated relationship as weights in a self-attention scheme (Wang et al., 2018). Zeng et al. (2021) used Neural Architecture Search (NAS) to search for an efficient network architecture (NPPNet) so as to tackle human parsing and pose estimation simultaneously. In their work, NPPNet adopted individual encoders and decoders for the two tasks, and NAS was used to search for both encoder and decoder structures as well as other module structures, including a multi-scale feature interaction module and a feature fusion module. Although NAS can help to find a better structure, as the network structure becomes more complex, the search space as well as the search time increase. *Instead, a shared encoder approach, focusing on solving the training imbalance problem caused by sharing is adopted in the present study.*

2.3. Pose estimation

The task of pose estimation aims to derive the coordinates of the different human joints from a human image. With the development of deep learning (DL) techniques, many DL-based methods have been proposed with promising results being achieved on standard benchmark datasets, such as MPII (Andriluka et al., 2014) and COCO (Lin et al., 2014). These methods can be categorized into regression-based and heatmap type based methods. Regression-based methods directly derive the human joint coordinates from input images with fewer intermediate stages and are easy to embed into end-to-end training networks. Since pose estimation requires the precise coordinate and location of the joints, this approach is difficult to optimize and hence regression-based methods always use a cascaded approach (Toshev and Szegedy, 2014; Carreira et al., 2016; Li et al., 2021a). Heatmap type based methods derive the heatmaps, on which each value indicates the probability of the joints occurring at the respective pixel location, and then decode the heatmaps to joint coordinates. Heatmaps represent 2D Gaussian distribution as centred on the actual joint coordinates. Compared to joint coordinates, heatmaps provide enriched characterization information and facilitate the regression optimization, thus heatmap type based pose estimation has received more attentions (Tompson et al., 2015; Newell et al., 2016). Newell et al. (2016) proposed a stacked hourglass network using repeated pooling, down- and up-sampling processes to derive the spatial distribution. Xiao et al. (2018) proposed a simplified baseline for human pose estimation and pose tracking by simply adding a few deconvolution layers in the final convolution stage. Wang et al. (2020) proposed an HRNet model, which is able to maintain the high-resolution representation at each stage of the network, and thus has become the state-of-the-art model for pose estimation.

2.4. Knowledge distillation

Knowledge distillation (KD) aims at creating a small "student" model, which learns from a large "teacher" model and was first introduced by Hinton et al. (2015). The large teacher model is used to supervise the small student model in order to achieve competitive or superior performance (Romero et al., 2014). KD methods can be classified in different ways. For example, according to where distillation takes place within the network, there are two classes of KD methods. One is to distil knowledge from the output of the teacher network to that of the student network (Chen et al., 2017a; Zhang et al., 2019; Ding et al., 2019). The other class mimics the features between the teacher and student networks (Komodakis and Zagoruyko, 2017; Heo et al., 2019; Tung and Mori, 2019; Yim et al., 2017). Alternatively, KD methods can also be classified into online or offline distillation. Offline distillation is more common that most existing KD methods adopt (Hinton et al., 2015; Romero et al., 2014; Komodakis and Zagoruyko, 2017; Mirzadeh et al., 2020) because of its simplicity and efficiency. A two-stage process is involved in the offline scheme, namely (1) a training phase of the teacher model and (2) a distillation phase which involves transferring knowledge from the teacher to the student network. Online distillation (Wang and Yoon, 2021) simplifies the training into a single stage, and all the networks are treated as students. Online distillation merges the training processes of all student networks, enabling them to gain extra knowledge from each other. In pursuit of effective and comprehensive knowledge transfer, Rao et al. (2023) introduced a parameter-efficient and student-friendly method for knowledge distillation. For pose estimation, Li et al. (2021b) employed an online scheme to distill assembled heatmaps of multiple branches. In federated learning, Zhang et al. (2022a) utilized KD technology to transfer knowledge from local models to global models. In non-autoregressive translation, KD technology is used to address challenges such as multiple lexical choices in the raw data (Ding et al., 2021). KD methods are typically applied to a single task optimization like Li et al. (2021b), whereas the current study focuses on knowledge distillation between multi-task branches in a MTL network. Within the context of MTL, Li and Bilen (2020) proposed a KD-based method to solve the imbalance problem. They first trained task-specific models, and then used task-specific adaptors to align features, enabling a balanced parameter sharing across tasks. However, this method did not consider the loss values of different tasks and it will fail when the loss values vary greatly between tasks.

3. The proposed method

This section introduces the proposed KD-based method for multi-task learning of human parsing and pose estimation. Fig. 1 shows the overall framework, which consists of a MTL model for human parsing and pose estimation (Fig. 1(b)), a task-specific model for human parsing (HP model of Fig. 1(a)) and a task-specific model for pose estimation (PE model of Fig. 1(c)). The models for single tasks are trained offline to guide the learning of the corresponding task in the MTL network. Both task-specific representations and logits are distilled from the single task-specific models and transferred to the multi-task model via the task-specific representation distillation (TRD) module and logit distillation (LD) module, respectively. In this paper, a cross-task representation consistency (CRC) module is also proposed to enhance the related information-sharing between the tasks, with the task-specific model-based loss weighting method being used to balance the loss values.

3.1. Problem formulation

Given an input image $I \in \mathbb{R}^{3\times\mu\times\lambda}$ of size $\mu \times \lambda$, the goal is to predict the label for each pixel $S \in \{1, 2, \ldots, L\}^{\mu\times\lambda}$ and the locations $P = (u_j, v_j)_{j=1}^{J}$ of the human body joints. Here (u_j, v_j) denote the coordinates of the jth joint, and J and $\mathcal{L}$ are the numbers of joints and human parsing classes, respectively. Each specific task is learned using an individual network, which consist of a backbone, an encoder for the task-specific representation learning and a prediction layer. Fig. 1(a) and (c) show the network structure of the task-specific human parsing (HP) model and that of task-specific pose estimation (PE) model, respectively.

The HRNetV2 is used as the backbone because it learns the mixed representation of multiple scales and has performed well on both semantic segmentation (Wang et al., 2020) and pose estimation (Wang et al., 2020). Following the backbone, an encoder module is used to extract features for the specific task. The encoder module is only a 1×1 convolution layer followed by BatchNorm and ReLU layers, because the focus in this paper is to investigate the balance between tasks and information exchange for mutual benefit. It is believe that better encoder for dense pixel prediction, such as OCR (Yuan et al., 2019), can also be utilized to enhance the representation learning. This will be discussed in later experimental Section 4.5. The feature extracted from the encoder of the HP model is denoted as $F_{ss} \in R^{c\times h\times w}$ and that from the encoder of the PE model as $F_{sp} \in R^{c\times h\times w}$, where c, h, w denote the channel number, height, and width of the feature map, respectively. Finally, the extracted feature is fed into a prediction layer to generate the logit prediction, which is denoted as $Z_{ss} \in R^{L\times h\times w}$ for the HP model and as $Z_{sp} \in R^{J\times h\times w}$ for the PE model.

The network structure of KDNet is shown in Fig. 1(b). It is similar to the task-specific model. The difference is that the HP and PE branches of KDNet share the same backbone. For KDNet, the extracted HP representation is denoted as $F_{ms} \in \mathbb{R}^{c\times h\times w}$ and the PE representation as $F_{mp} \in \mathbb{R}^{c\times h\times w}$. Then F_{ms} and F_{mp} are fed into a prediction layer to output the HP prediction $Z_{ms} \in R^{L\times h\times w}$ and $Z_{mp} \in R^{J\times h\times w}$, respectively.

3.2. Task-specific representation distillation (TRD) module

The task-specific representation distillation (TRD) module is designed to transfer the knowledge of the task-specific model to the KDNet. To enable KDNet learn discriminant representations of human parsing and pose estimation, the TRD module is added between the specific representations, instead of the shared representations in Li and Bilen (2020). Based on the knowledge distillation technology, the task-specific model is regarded as the teacher model and the KDNet as the student model, and the knowledge of the teacher model is then distilled to the student model. Specifically, the HP model and PE model are trained separately offline. After this, task-specific representations, F_{ss} and F_{sp}, are extracted and a representation distillation loss $\mathcal{L}_{F_{ss}}$ is added between F_{ss} and F_{ms}, and a loss $\mathcal{L}_{F_{sp}}$ is added between F_{sp} and F_{mp} to let F_{ms} mimic F_{ss}, and F_{mp} mimic F_{sp}. In order to match the feature map size of the task-specific representations, a transformation operation is applied to the extracted representation of the KDNet. The transformation operation is implemented by means of a 1×1 convolution followed by BatchNorm and ReLU layers. For the task-specific representation and the MTL representation, the TRD module first computes their pixel-wise correlation matrices, because semantic segmentation and pose estimation are both pixel-wise prediction and have similar structure. The formula to calculate the pixel-wise correlation matrices is as follows:

$$R_{ij}(a) = (\frac{a_i}{\|a_i\|_2})^T(\frac{a_j}{\|a_j\|_2}) \tag{1}$$

where subscripts i and j denote the position of the specific pixels, a_i is the ith feature representation of feature a at the ith pixel, and a_j is the jth feature representation of feature a at the jth pixel. To lower the memory consumption, the feature representations are downsampled to $\frac{1}{8}$ in size in the pixel-wise correlation calculation.

Lastly, the Euclidean distance is selected to measure the distance between the pixel-wise correlation matrix of the task-specific representation and that of the MTL representation, taking advantage of its strong capability of capturing geometric differences in the feature space. The overall TRD loss is computed by

$$\mathcal{L}^{TRD} = \mathcal{L}_2(R(\varphi_{ss}(F_{ss})), R(\varphi_{ms}(F_{ms}))) + \mathcal{L}_2(R(\varphi_{sp}(F_{sp})), R(\varphi_{mp}(F_{pm}))) \tag{2}$$

where φ_{ss} and φ_{sp} represent the transform operations and $\mathcal{L}_2$ is the Euclidean distance function between the L2 normalized features:

$$\mathcal{L}_2(a,b) = \| \frac{a}{\|a\|_2} - \frac{b}{\|b\|_2} \| \tag{3}$$

3.3. Logit distillation (LD) module

Since the output logit of the trained model encodes not only the soft labels but also the knowledge (e.g. the edge information) learned from the dataset, providing more accurate learning supervision, a logit distillation (LD) loss is added between the output of the task-specific model and that of the KDNet.

For the human parsing, the Kullback–Leibler (KL) divergence loss is used to match the soft output Q_{ss} of the task-specific HP model and the parsing output of the KDNet. For pose estimation, the mean square error (MSE) or $\mathcal{L}_2$ loss function is selected to measure the divergence between the pose prediction of the task-specific PE model (i.e. Z_{sp}) and that of the KDNet (i.e. Z_{mp}). The overall LD loss is computed by:

$$\mathcal{L}^{LD} = \mathcal{L}_{KL}(Z_{ms}, Z_{ss}) + \mathcal{L}_2(Z_{mp}, Z_{sp}) = T^2 \sum_{i,j} \sum_{l} Q_{ms} log \frac{Q_{ss}}{\hat{Q}_{ss}} + \|Z_{mp} - Z_{sp}\|_2^2 \tag{4}$$

where $$Q = \frac{exp(Z/T)}{\sum_l exp(Z/T)} \tag{5}$$

and T is the temperature parameter of KD (Hinton et al., 2015). The higher T, the smoother the output probability distribution, the information carried by the negative label will be relatively amplified, and the model training will pay more attention to the negative label. It is set as $T = 1$ in this paper.

3.4. Cross-task representation consistency (CRC) module

In order to strength the knowledge transfer between different tasks, a cross-task representation consistency (CRC) module is designed between the HP representation and PE representation. This module also realize the knowledge transfer between tasks, while the knowledge of the corresponding representation between single task model and MTL is done via the TRD modules described in Section 3.2. Leveraging the close relationship between human parsing and pose estimation, the CRC module facilitates knowledge transfer by aligning the consistency of their representations. Similar to TRD module, the pixel-wise correlation matrix of the HP representations is used to guide the learning of pose estimation, because the pixel-wise correlation matrices of HP and PE are similar while the HP representation has more semantic information. This will be further discussed later in experimental Section 4.4.2. The CRC module then uses the $\mathcal{L}_1$ distance function to align the pixel-wise correlation matrices R_{ms} and R_{mp} as follows:

$$\mathcal{L}^{CRC} = \mathcal{L}_1(R(f_{ms}), R(\tau(f_{mp}))) \tag{6}$$

where τ is the transformation function implemented via a 1×1 convolution followed by BatchNorm and ReLU layers.

This alignment ensures that the semantic information captured in the HP (Human Parsing) representation is effectively transferred to the PE (Pose Estimation) representation, thereby enhancing the learning process of both tasks.

3.5. Task-specific model-based loss weighting method

Different from previous work that equal weighting is used as the initial weights for MTL network training, a task-specific model-based loss weighting method is introduced here to fully leverage the variation of loss in single task model training to determine the corresponding initial loss weights for HP and PE tasks in the MTL model training. It is noted that the variation of loss reflects the progress of model learning. Fig. 2 shows the changes in training and validation losses, and the accuracy for the respective single HP model (in terms of the Mean IoU) and PE model (in terms of PCK) along the training process. As can be seen from the figure, they have different learning speeds and loss values. Let s denotes the epoch when the HP model obtains best mean IoU and p denotes the epoch when the PE model reaches the best PCK on the validation dataset, and they are respectively computed by:

$$s = \underset{i<epochs}{argmax}(IoU_i) \tag{7}$$

$$p = \underset{i<epochs}{argmax}(PCK_i) \tag{8}$$

where $epcohs$ is the number of training epochs.

Due to the mutual influence between tasks, the learning speed of KDNet may change considerably compared to the single-task training, which makes it difficult to set appropriate weights between tasks for MTL training. Nevertheless, the best result achieved by a single-task model is used as a reference lower bound for a multi-task model in the current method. Hence, when the training of single-task models converge, namely at epoch s and p obtained in Eqs. (7) and (8), the specific loss values are used to assign *weights* for MTL model training in order to balance the loss scales. Let L_{ss}^s denotes the training loss of HP model at the sth epoch and L_{sp}^p be the training loss of PE model at the pth epoch, then the product of the task loss and its weight should be equal, namely

$$\omega_s \mathcal{L}_{ss}^s = \omega_p \mathcal{L}_{sp}^p \tag{9}$$

The HP loss weight ω_s and PE loss weight ω_p are respectively computed as follows:

$$\omega_s = \frac{\mathcal{L}_{ss}^s + \mathcal{L}_{sp}^p}{\mathcal{L}_{ss}^s} \tag{10a}$$

$$\omega_p = \frac{\mathcal{L}_{ss}^s + \mathcal{L}_{sp}^p}{\mathcal{L}_{sp}^p} \tag{10b}$$

These loss weights ω_s and ω_p are then used as the loss weights to optimize the network during the training.

3.6. Loss function

The overall loss function shown in Eq. (11) below consists of a cross entropy loss for HP, a MSE loss for PE, task-specific representation matching loss, logit distillation loss and cross-task representation consistency loss.

$$\mathcal{L} = \omega_s \mathcal{L}_{ms} + \omega_p \mathcal{L}_{mp} + \mathcal{L}^{TRD} + \mathcal{L}^{LD} + \mathcal{L}^{CRC} \tag{11}$$

where $\mathcal{L}_{ms}$ is the cross-entropy loss between the parsing results Z_{ms} and the parsing annotations S, $\mathcal{L}_{mp}$ is the MSE loss between the joint predictions Z_{mp} and the true joint locations P. The joint locations are encoded as heatmaps as they are beneficial to preserve spatial information and are widely used in pose estimation. For encoding joint location, the distribution-based method (Zhang et al., 2020a), rather than the standard heatmap encoding (Tompson et al., 2015), is used because the former encoding scheme is more accurate and bias-free (Zhang et al., 2020a).

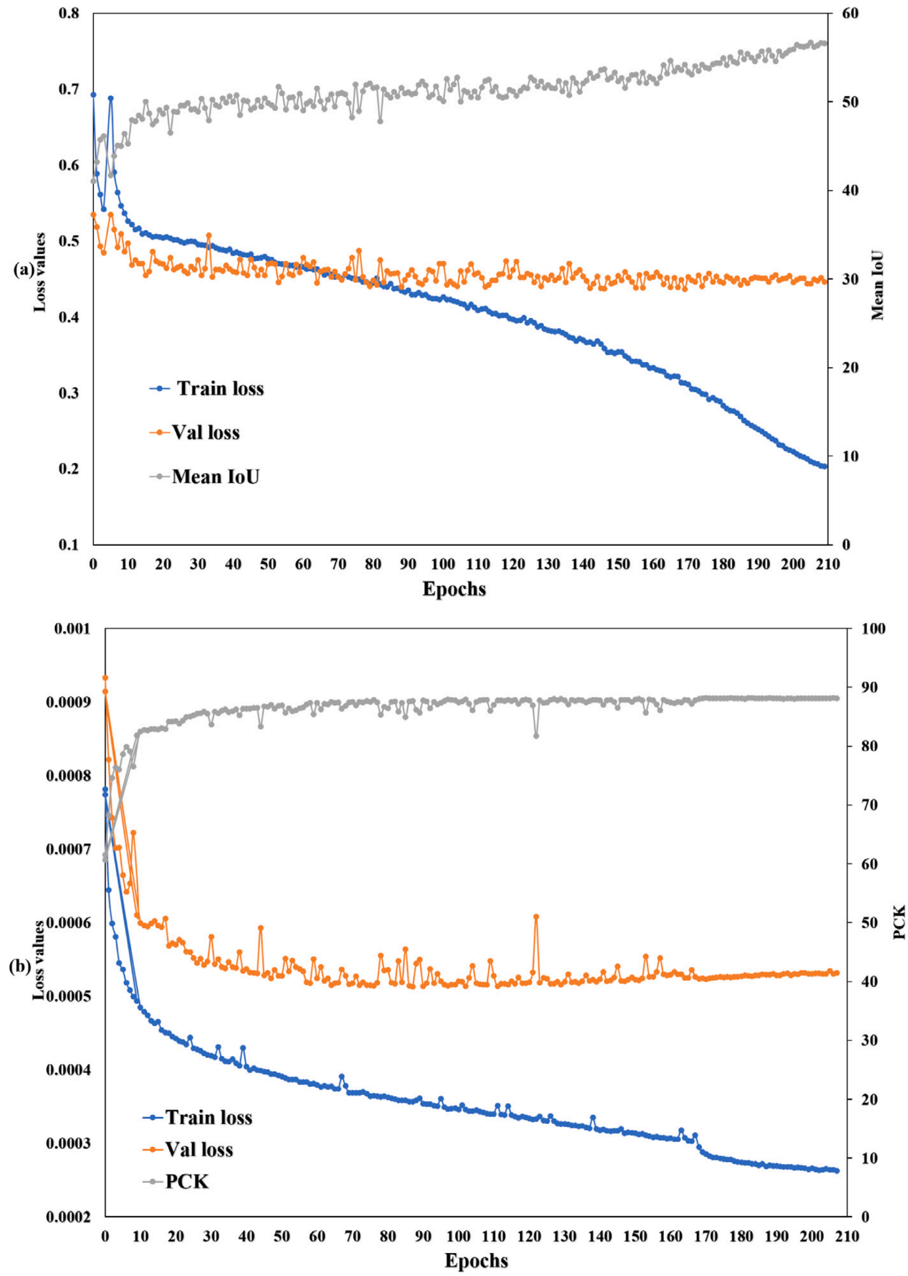

Fig. 2. (a) The variation in the training loss, validation loss and the Mean IoU of the single HP model based on the LIP dataset; (b) the variation in the training loss, validation loss and PCK of the single PE model based on the LIP dataset.

4. Experimental results and discussions

To evaluate the effectiveness of the proposed KDNet, detailed experiments on three datasets were carried out. In the following subsections, these datasets are described as well as the metrics for evaluation, the details of the implementation, and the results obtained here compared with those of the other methods. An ablation study was also carried out on the LIP dataset for both the HP and PE tasks.

4.1. Datasets

The LIP dataset (Liang et al., 2018) is a large benchmark dataset annotated with segmentation of both the human body and the clothing regions as well as of the key joint locations. It contains 50,462 images, each of which has annotated location coordinates of 16 keypoints of the body and pixel-level labels of 20 body part/clothing classes. The dataset is partitioned into a training set (30,462 images), validation set (10,000 images) and test set (10,000 images).

The extended Pascal-Person-Part (ePPP) dataset (Liang et al., 2016) is a dataset with 3533 multiple-person images, each annotated with 6 human body parts and 14 joints. Since the dataset is a multi-person dataset, "Detectron2" (Wu et al., 2019) was used here to generate single-person images. The dataset is partitioned into a training set of 1716 images and testing set of 1817 images.

The MPII dataset (Andriluka et al., 2014) is a large-scale and state-of-the-art benchmark for human pose estimation. There are 40,000 human images, of which 28,000 images are used for training and 11,000 images are used for validation. It annotates 16 body joints, including head top, upper neck, thorax, pelvis, left/right shoulder, left/right hip, left/right elbow, left/right wrist, left/right knee, and left/right ankle.

Table 1
The metrics used in evaluation of different datasets.

Dataset	Accuracy metrics				Efficiency metrics			
	Human parsing			Pose estimation	Human parsing & Pose estimation			
LIP	Mean IoU↑	Pixel acc.↑	Mean acc.↑	PCK↑	Params↓	FLOPs↓	Training time↓	Inference time↓
ePPP	Mean IoU↑	Pixel acc.↑	Mean acc.↑	PCK↑	–	–	–	–
MPII				mAP↑	–	–	–	–

4.2. Metrics and implementation details

In terms of *human parsing*, the pixel accuracy, mean accuracy, and mean accuracy on the intersection over the union region (IoU) were used for evaluation based on the LIP and ePPP datasets, because these metrics are widely used in segmentation evaluations. To evaluate the accuracy of *pose estimation*, following previous work, the Percentage of Correct Keypoints (PCK) was used for LIP dataset and MPII datasets while the Mean Average Precision (mAP) was used for the ePPP dataset. The PCK is a metric to measure the percentage of correctly predicted keypoints relative to a ground truth, which is commonly used in the field of human pose estimation to assess the precision of joint localization. We calculate the percentage of correct keypoints within the threshold 0.5 of the head length (PCK@0.5). The mAP calculates the average precision across multiple keypoints, providing a comprehensive measure of model accuracy and robustness across different scenarios. The mAP is computed as follows:

$$mAP = meanAP_{(0.50:0.05:0.95)} \tag{12}$$

$$AP_s = \frac{\sum_p \delta(OKS_p > s)}{\sum_p 1} \tag{13}$$

$$OKS_p = \frac{\sum_i exp(-d_{pi}^2/(2a_p^2\theta_i^2))\delta(v_{pi} > 0)}{\sum_i \delta(v_{pi} > 0)} \tag{14}$$

where p is the person instance index, i is the keypoint index, $\delta()$ is the Kronecker function, d_{pi} is the Euclidean distance between the ith predicted joint of the pth person and its ground truth, a_p is the occupied area of the pth person, θ_i is the normalization factor predefined for each keypoint type, and v_{pi} is the visible status. To evaluate the efficiency of our method, we utilized the model parameters, FLOPs, training time and inference time. The training time is the average time taken per batch during training, while the inference time indicates the average time processing per image during inference. We only compared the efficiency performance on the LIP dataset, because they are almost the same on all datasets. Table 1 shows the metrics used on different datasets.

All the experiments were conducted on PyTorch. The input size of each image was 384 × 384 for the LIP dataset and the ePPP dataset and 256 × 256 for the MPII dataset, and the training images were augmented by mirroring, random scaling in $[0.5, 1.5]$, random rotation in $[-40, 40]$ and normalization. In the LIP and ePPP datasets, the HP and PE single task models were trained first, and then the KDnet was trained by distilling knowledge from HP and PE models. As MPII dataset has no annotations of human parsing, the HP model used was trained on the LIP dataset. The network was trained for 210 epochs using a pre-trained HRNet-W48 model on ImageNet dataset, because HRNet (Sun et al., 2019) is a commonly chosen baseline for most of state-of-the-art models of human parsing (Liu et al., 2020; Zhang et al., 2022c; Yang et al., 2022) and pose estimation (Wang et al., 2020; Zhang et al., 2020a, 2022c), and the experiment were conducted using two GPUs on A100. All the models were trained using Adam optimizer with an initial learning rate of 0.001, which was dropped by one-tenth at the 170th and 200th epochs. For a fair comparison with single task models, in the inference phase, flipping and multi-scale with an interval in $[0.75, 1.0, 1.25]$ were used for the human parsing task while only flipping was used for pose estimation. All human parsing results were resized to the original size of the image and the predicted human joint coordinates were transformed back to the original coordinate space of the image for performance evaluation.

4.3. Experimental results

4.3.1. Results on the LIP dataset

Table 2 compares the **human parsing** results of the proposed KDNet with those of the state-of-the-art methods based on the LIP validation set. Table 3 shows per-class accuracy comparison. As can be seen, the new method of KDNet achieves the highest mean IoU (60.16%) of all the listed state-of-the-art methods. CorrPM (Zhang et al., 2020c) and NPPNet (Zeng et al., 2021) are the recent approaches that fuse human parsing and pose estimation to assist each other. NPPNet (Zeng et al., 2021) is the only method that searches an efficient network architecture to obtain consistent performance for both tasks. Compared with NPPNet (Zeng et al., 2021), the proposed method exceeds by 1.6% in terms of mean IoU. Moreover, NPPNet (Zeng et al., 2021) is search based technology and is not flexible when applying to other backbones, while the proposed method can easily work with different backbones, encoders and decoders. CorrPM (Zhang et al., 2020c) learns the correlation among pose, parsing and edge features so as to improve human parsing or pose estimation with a self-attention based module. This approach introduces extra model parameters and thus increases computation complexity. Comparatively, the proposed KDNet utilizes knowledge distillation to capture the relationship between human parsing and pose estimation, and improves the performance by 4.83% (60.16%–55.33%) without increasing model parameters and computation complexity in the inference phase. Compared to HTCorrPM (Zhang et al., 2022c), which is the CorrPM model based on HRNetV2-W48 (Wang et al., 2020), the proposed KDNet still outperforms by 2.39%, even though the edge information is utilized in both HTCorrPM and CorrPM but not in the current KDNet. On the other hand, the proposed KDNet also performs better than hierarchical methods which leverage the hierarchy structure of the human body and iteratively fuse the information over the human structure, as measured by Mean IoU. Although one method of HHP (Wang et al., 2021) achieved a slightly higher pixel accuracy (88.94% vs. 89.05%) compared to our KDNet, it involved a large number of model parameters and demand on high computational resources. For example, the size of HHP is 268.59 MB, which is more than double the size of KDNet. (This will be further discussed in Section 4.5.1 and Table 11 later.) DCGNet (Liu et al., 2022) and QANet (Yang et al., 2022) are the most recent approaches for human parsing. Based on the ResNet-101 backbone, DCGNet (Liu et al., 2022) significantly increases the performance to 59.04%, which is only 0.2% lower than KDNet. DCGNet (Liu et al., 2022) is a completely different approach with more complex network design and is potentially complementary to the current method. In addition to accurate predictions, KDNet also performs well in efficiency, which will be discussed further in Section 4.5.1. Compared to QANet (Yang et al., 2022) with the HRNet-W48 backbone, KDNet surpasses by 0.55% in terms of mean IoU.

Table 4 compares the **pose estimation** results based on the LIP validation dataset, involving two pose-only models (Wang et al., 2020; Osokin, 2019) as well as four models (Liang et al., 2018; Nie et al., 2018a; Zhang et al., 2022c; Zeng et al., 2021) use human parsing for pose estimation. It can be seen that the present new model KDNet outperforms all the listed models and yields 88.6% in terms of PCK. GCM (Osokin, 2019), a hourglass-based network, uses a U-shaped context encoder module for pose estimation. This method is complementary to the proposed method, and may further improve the performance. JPPNet (Liang et al., 2018) and HTCorrPM (Zhang et al.,

Table 2
Human parsing result comparison in terms of overall accuracy based on the validation set of the LIP dataset.

Methods	Input size	Edge	Pose	Pixel acc.↑	Mean acc.↑	Mean IoU↑
MMAN (ECCV18) (Luo et al., 2018)	256 × 256			–	–	46.81
MuLA (ECCV18) (Nie et al., 2018a)[a]	256 × 256		✓	88.50	60.50	49.30
SS-NAN (CVPR17) (Zhao et al., 2017b)	321 × 321			87.59	56.03	47.92
JPPNet (PAMI18) (Liang et al., 2018)	384 × 384		✓	86.39	62.32	51.37
CE2P (AAAI19) (Ruan et al., 2019)[a]	473 × 473	✓		87.37	63.20	53.10
SLRS (CVPR20) (Li et al., 2020b)[a]	473 × 473	✓		88.33	66.53	54.12
BraidNet (MM19) (Liu et al., 2019b)[a]	384 × 384			87.60	66.09	54.40
CorrPM (CVPR20) (Zhang et al., 2020c)[a]	384 × 384	✓	✓	87.68	67.21	55.33
HTCorrM (PAMI22) (Zhang et al., 2022c)[a]	384 × 384	✓	✓	–	–	56.85
HTCorrM (PAMI22) (Zhang et al., 2022c)[b]	384 × 384	✓	✓	88.10	70.95	57.86
OCR (ECCV20) (Yuan et al., 2019)[a]	473 × 473			–	–	55.60
PCNet (CVPR20) (Zhang et al., 2020b)	473 × 473			–	–	57.03
CNIF (ICCV19) (Wang et al., 2019)	473 × 473			88.03	68.80	57.74
PRM (TMM22) (Zhang et al., 2022b)	473 × 473			–	–	58.86
HHP (CVPR20) (Wang et al., 2021)	473 × 473			**89.05**	70.58	59.25
NPPNet(ICCV2021) (Zeng et al., 2021)	384 × 384	✓	✓	–	–	58.56
SCHP (PAMI20) (Li et al., 2020a)	473 × 473	✓		–	–	59.36
DTCF (Liu et al., 2020)	473 × 473	✓		88.61	68.89	57.82
CDGNet (Liu et al., 2022)[a]	473 × 473	✓		88.53	70.89	59.04
QANet (TMM22) (Yang et al., 2022)	512 × 384			88.92	71.17	59.61
KDNet (this study)[a]	384 × 384		✓	88.45	71.17	59.24
KDNet (this study)	384 × 384		✓	88.94	**71.47**	**60.16**

[a] Means single-scale testing results without flipping.
[b] Means with loss weights calculated by the proposed method in the present study.

Table 3
Human parsing result comparison in terms of per-class Mean IoU based on the validation set of the LIP dataset.

Methods	Hat	Hair	Glove	Glass	U-cloth	Dress	Coat	Sock	Pants	Jumpsuit	Scarf
MMAN (ECCV18) (Luo et al., 2018)	57.66	65.63	30.07	20.02	64.15	28.39	51.98	41.46	71.03	23.61	9.65
SS-NAN (CVPR17) (Zhao et al., 2017b)	63.86	70.12	30.63	23.92	70.27	33.51	56.75	40.18	72.19	27.68	16.98
JPPNet (PAMI18) (Liang et al., 2018)	63.55	70.20	36.16	23.48	68.15	31.42	55.65	44.56	72.19	28.39	18.76
CE2P (AAAI19) (Ruan et al., 2019)[a]	65.29	72.54	39.09	32.73	69.46	32.52	56.28	49.67	74.11	27.23	14.19
SLRS (CVPR20) (Li et al., 2020b)[a]	65.74	71.55	42.58	30.62	69.44	37.13	56.05	47.34	74.92	31.18	23.77
BraidNet (MM19) (Liu et al., 2019b)[a]	66.80	72.00	42.50	32.10	69.80	33.70	57.40	49.00	74.90	32.40	19.30
CorrPM (CVPR20) (Zhang et al., 2020c)[a]	66.20	71.56	41.06	31.09	70.20	37.74	57.95	48.40	75.19	32.37	23.79
OCR (ECCV20) (Yuan et al., 2019)[a]	67.42	73.16	44.01	36.31	69.90	35.53	57.03	51.87	75.09	29.85	19.16
PCNet (CVPR20) (Zhang et al., 2020b)	69.32	73.08	44.72	34.21	72.59	36.02	60.84	51.03	76.66	38.78	31.60
CNIF (ICCV19) (Wang et al., 2019)	69.55	73.45	45.17	41.45	70.57	38.52	57.94	54.02	75.07	28.00	31.92
NPPNet (ICCV21) (Zeng et al., 2021)	66.43	72.34	**51.98**	31.59	71.88	**40.88**	60.54	49.81	77.07	27.55	26.58
PRM (TMM22) (Zhang et al., 2022b)	**70.83**	**74.22**	50.32	40.02	**73.12**	38.52	**63.81**	53.64	**78.51**	**40.05**	**32.51**
SCHP (PAMI20) (Li et al., 2020a)	70.63	74.09	51.40	**41.70**	70.56	40.06	58.17	**55.17**	76.57	33.78	26.63
KDNet (this study)	70.32	74.16	49.84	40.79	71.56	38.90	59.26	54.66	78.00	31.06	30.60

Methods	Skirt	Face	L-arm	R-arm	L-leg	R-leg	L-shoe	R-shoe	Bkg	Ave.
MMAN (ECCV18) (Luo et al., 2018)	23.20	69.54	55.30	58.13	51.90	52.17	38.58	39.05	84.75	46.81
SS-NAN (CVPR17) (Zhao et al., 2017b)	26.41	75.33	55.24	58.93	44.01	41.87	29.15	32.64	88.67	47.92
JPPNet (PAMI18) (Liang et al., 2018)	25.14	73.36	61.97	63.88	58.21	57.99	44.02	44.09	86.26	51.37
CE2P (AAAI19) (Ruan et al., 2019)[a]	22.51	75.5	65.14	66.59	60.10	58.59	46.63	46.12	87.67	53.10
SLRS (CVPR20) (Li et al., 2020b)[a]	30.44	74.70	64.73	67.27	57.18	57.89	45.82	46.30	87.70	54.12
BraidNet (MM19) (Liu et al., 2019b)[a]	27.20	74.90	65.50	67.90	60.20	59.60	47.40	47.90	88.00	54.40
CorrPM (CVPR20) (Zhang et al., 2020c)[a]	29.23	74.36	66.53	68.61	62.80	62.81	49.03	49.82	87.77	55.33
OCR (ECCV20) (Yuan et al., 2019)[a]	26.63	75.95	66.81	68.65	63.23	62.12	50.60	50.68	87.96	55.60
PCNet (CVPR20) (Zhang et al., 2020b)	33.94	76.65	67.07	68.74	60.22	60.16	47.65	48.67	88.68	57.03
CNIF (ICCV19) (Wang et al., 2019)	30.20	76.38	68.28	69.49	65.52	65.51	52.67	53.38	87.99	57.74
NPPNet (ICCV21) (Zeng et al., 2021)	33.54	75.31	71.04	71.50	**71.75**	70.55	56.66	55.84	88.38	58.56
PRM (TMM22) (Zhang et al., 2022b)	**35.13**	**77.26**	68.52	68.92	62.32	61.35	49.64	49.75	88.82	58.86
SCHP (PAMI20) (Li et al., 2020a)	32.83	76.63	69.33	71.76	67.93	67.42	56.56	57.55	88.40	59.36
KDNet (this study)	32.63	77.15	**72.14**	**73.57**	71.63	**71.01**	**58.21**	**58.79**	**89.01**	**60.16**

[a] Means single-scale testing results without flipping.

2022c) train the single task model individually, whereas KDNet simultaneously optimizes both tasks in a MTL network, outperforming JPPNet (Liang et al., 2018) by 6.1% and HTCorrPM (Zhang et al., 2022c) by 1.6% in pose estimation performance. MuLA (Nie et al., 2018a) and NPPNet (Zeng et al., 2021) use the separated networks to learn representation of different tasks, while the proposed KDNet shares feature representations and model parameters, lowering the risk of overfitting. MuLA (Nie et al., 2018a) uses preliminary representations for mutual guidance and introduces a mutual adaptation module to dynamically predict model parameters. Instead, KDNet augments representation for both tasks by aligning the similar structure feature between tasks, outperforming MuLA by 3.2%. Compared to NPPNet (Zeng et al., 2021), KDNet employs hard parameter sharing and has fewer model parameters and lower complexity. Taking advantage of the correlation between tasks, KDNet designs simple yet effective information exchange modules, saving extra time for architecture search, achieving a 0.6% performance gain than Zeng et al. (2021). HRNet (Sun

Table 4
Pose estimation result comparison based on the validation set of the LIP dataset.

Method	Head	Sho.	Elb.	Wri.	Hip	Knee	Ank.	PCK↑
JPPNet (Liang et al., 2018)	93.2	89.3	84.6	82.2	69.9	78.0	77.3	82.5
GCM (Osokin, 2019)	93.7	90.4	85.5	82.1	73.4	80.2	80.2	84.0
MuLA (Nie et al., 2018a)	–	–	–	–	–	–	–	85.4
HTCorrM (Zhang et al., 2022c)	–	–	–	–	–	–	–	87.0
NPPNet (Zeng et al., 2021)	–	–	–	–	–	–	–	88.2
HRNet (Wang et al., 2020)	94.7	93.2	88.7	87.1	**78.5**	85.9	**86.3**	88.0
KDNet	**95.7**	**93.4**	**90.4**	**88.3**	77.7	**86.6**	**86.3**	**88.6**

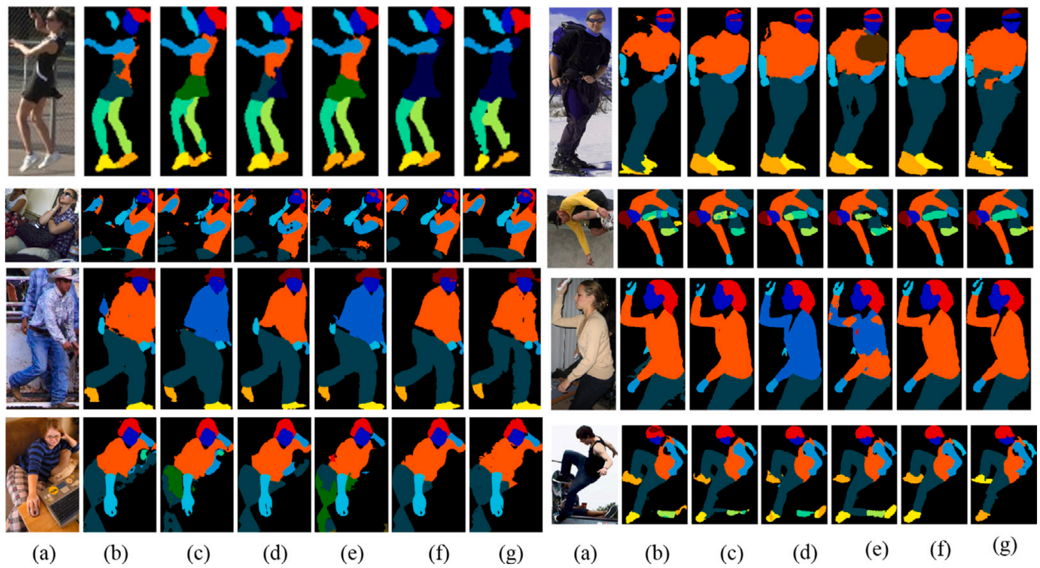

Fig. 3. Qualitative comparison of human parsing results: (a) input images, results of (b) CE2P (Ruan et al., 2019), (c) CorrPM (Zhang et al., 2020c), (d) OCR (Yuan et al., 2020), (e) SCHP (Li et al., 2020a) and (f) KDNet (current method), and (g) ground truth.

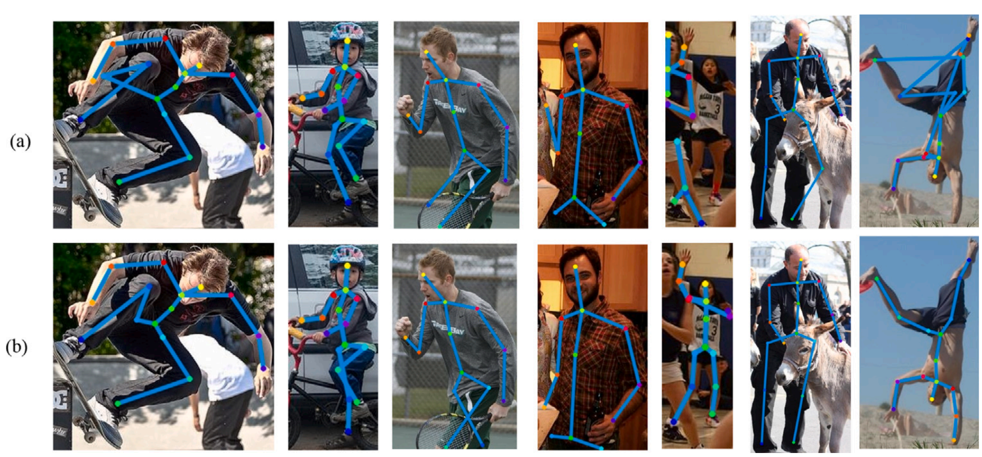

Fig. 4. Qualitative comparison of pose estimation results: (a) single PE model and (b) our results.

et al., 2019) is the baseline of the present KDNet model. With the proposed knowledge distillation strategy for balancing multi-task learning, the performance of KDNet is increased by 0.6%.

Qualitative results of human parsing Fig. 3 shows the qualitative results of human parsing on the LIP dataset. The results of KDNet are compare with that of four state-of-the-art methods, namely CE2P (Ruan et al., 2019), CorrPM (Zhang et al., 2020c), OCR (Yuan et al., 2020) and SCHP (Li et al., 2020a). Observed from the second row, it can seen that KDNet correctly segments the clothing and body regions even with strange camera views. For images with complex human poses and/or severe occlusions, KDNet still performs much better than other methods, as shown in the last row. It demonstrates that KDNet has robust human parsing performance with the help of distilled knowledge from pose estimation.

Qualitative results of pose estimation Fig. 4 shows the qualitative results of pose estimation on the LIP dataset. It compares the results of KDNet (MTL model) with the single PE model. As shown, KDNet obtains better prediction results than the single PE model in condition with diverse postures, complex background and severe occlusions. For example, in the last column of Fig. 4, the person in the image is upside down. With such an input image, the single PE model fails because of confusion about the upper and lower limb joints, whereas KDNet accurately locates all the joint points.

Table 5
Pose estimation result comparison based on the testing set of the ePPP dataset.

Method	Head	Sho.	Elb.	Wri.	Hip	Knee	Ank.	mAP↑
Xia et al. (2017)	58.0	52.1	43.1	37.2	22.1	30.8	31.1	39.2
PIL (Nie et al., 2018b)	67.8	56.6	45.7	41.9	24.2	26.4	24.2	41.0
MuLA (Nie et al., 2018a)	–	–	–	–	–	–	–	39.9
NPPNet (Zeng et al., 2021)	**68.1**	57.9	46.1	42.0	27.3	27.6	25.7	42.1
KDNet	62.0	**61.3**	**54.6**	**56.0**	**28.0**	**36.2**	**35.4**	**47.6**

Table 6
Human parsing result comparison in terms of per-class Mean IoU based on the testing set of the ePPP dataset.

Methods	Pose	Head	Torso	U-Arm	L-Arm	U-Leg	L-Leg	Bkg	**Ave.**
Attention (CVPR16) (Chen et al., 2016)		81.47	59.06	44.15	42.50	38.28	35.62	93.65	56.39
JPPNet (PAMI18) (Liang et al., 2018)	✓	83.26	62.40	47.80	45.58	42.23	39.48	94.68	59.36
Attention + MMAN (ECCV18) (Luo et al., 2018) [†]		82.58	62.83	48.49	47.37	42.80	40.40	94.92	59.91
Joint (Xia et al., 2015)	✓	85.50	67.87	54.72	54.30	48.25	44.76	95.32	64.69
DeepLabV2 (PAMI2017) (Chen et al., 2017b)		–	–	–	–	–	–	–	64.94
MuLA (ECCV18) (Nie et al., 2018a)	✓	84.60	68.30	57.50	54.10	49.60	46.40	95.60	65.10
DeepLabV3+ (ECCV18) (Chen et al., 2018a)		87.02	72.02	60.37	57.36	53.54	48.52	96.07	67.84
WSHP (CVPR18) (Fang et al., 2018)	✓	87.15	72.28	57.07	56.21	52.43	50.36	97.72	67.60
SPGNet (ICCV19) (Cheng et al., 2019)		87.67	71.41	61.69	60.35	52.62	48.80	95.98	68.36
PGN (ECCV18) (Gong et al., 2018)		**90.89**	75.12	55.83	64.61	55.42	41.57	95.33	68.40
CNIF (ICCV19) (Wang et al., 2019)		88.02	72.91	64.31	63.52	55.61	54.96	96.02	70.76
SCHP (PAMI20) (Li et al., 2020a)		87.41	73.80	64.98	64.70	57.43	55.62	96.26	71.46
DTCF (MM20) (Liu et al., 2020)		88.32	73.54	64.19	63.91	55.01	54.34	96.25	70.80
HHP (CVPR20) (Wang et al., 2021)		89.73	**75.22**	66.87	66.21	58.69	58.17	96.94	**73.12**
NPPNet (ICCV21) (Zeng et al., 2021)	✓	86.78	72.50	66.33	63.95	58.06	58.59	95.88	71.73
KDNet (current)	✓	86.81	74.18	67.30	65.96	57.20	56.82	95.91	72.03
KDNet + edge	✓	87.05	74.28	**67.64**	**66.68**	56.95	57.59	95.98	72.31
KDNet – OCR	✓	87.95	74.79	67.33	66.48	58.49	59.08	**96.31**	72.92
KDNet – OCR w/o gt	✓	88.02	74.62	67.28	66.57	**58.65**	**59.19**	96.20	72.93

4.3.2. Results on the ePPP dataset

Tables 5 and 6 compare various methods in terms of their performance in pose estimation and in human parsing on the ePPP testing dataset. It can be observed that the performance of KDNet method is remarkable for both tasks. For **pose estimation**, the present method surpasses (Xia et al., 2017), PIL (Nie et al., 2018b) and MuLA (Nie et al., 2018a) by more than 8.2%, 6.6% and 7.7% in terms of mAP. For **human parsing**, the present method outperforms Attention (Chen et al., 2016), JPPNet (Liang et al., 2018) and attention+MMAN (Luo et al., 2018) by over 10% in mean IoU. In particularly, the mAP and mean IoU values of the present method are respectively 5.5% higher (47.6% – 42.1%) and 0.3% higher (72.03% – 71.73%) than that of NPPNet (Zeng et al., 2021). This is because that KDNet facilitates communication not only between specific-task models and the MTL model but also between specific-task branches within the MTL model through the proposed TRD, LD, and CRC modules. In contrast, NPPNet only supports communication between specific-task branches of the MTL model. Another reason is that KDNet takes into account task-specific model losses to balance the loss levels, whereas NPPNet uses Uncertainty loss, which fails to address the issue due to the significant imbalance in loss levels between parsing and pose estimation.

4.3.3. Results on the MPII dataset

Table 7 shows the pose estimation comparison results on the validation set of the MPII dataset. As shown, benefiting from the distilled knowledge from human parsing, KDNet obtains the highest accuracy in almost all joints, achieving an overall PCK of 91.0%. As MPII dataset has no parsing annotations, thus the cross entropy loss $\mathcal{L}_{ms}$ for HP (Eq. (11)) is not used in the training. However, the HP model distilled from single HP model trained on other dataset such as ePPP still contains much information about human parsing, like the intermediate representation and parsing prediction, which can be transferred to pose estimation through the other losses, such as task-specific representation matching loss, logit distillation loss and cross-task representation consistency loss. It shows that KDNet can be deployed on dataset missing ground truth labels for one of the two tasks, and can still achieve state-of-the-art performance. Table 7 has demonstrated the effectiveness and adaptability of the proposed KDNet. Compared to Hybrid-Pose (Kamel et al., 2020) that uses two ConvNet models to refine pose estimation, the proposed KDNet outperforms it by 0.6% of PCK. It is interesting to note that even though HTCorrM (Zhang et al., 2022c) uses both parsing and edge to assist pose estimation, KDNet still outperforms it by 0.3%. The reason is that KDNet transfers more knowledge to the MTL model, including knowledge from both HP and PE models, compared to HTCorrM.

4.4. Ablation study

Ablation study was conducted on the LIP dataset, with experiments being performed to evaluate the effectiveness of (1) the proposed loss weighting algorithm, (2) cross-task representation consistency module, and (3) task-specific knowledge distillation module. In the ablation studies, multi-scale testing and flipping were not applied.

4.4.1. Comparison of different loss weights

The performance of the present loss weighting method (see Section 3.5 and Eq. (10)) is compared to single task model, equal loss weighting, setting loss weight based on degree of uncertainty (Kendall et al., 2018) and other weight setting of KD-based MTL (Li and Bilen, 2020) in Table 8. As shown, the present loss weighting method balances better between different tasks than the other loss weighting methods. Equal loss weighting is totally biased towards the learning of human parsing and performs poorly in terms of pose estimation. The uncertainty-based (Kendall et al., 2018) and KD-based (Li and Bilen, 2020) methods alleviate this problem to some extent, but their performance in terms of pose estimation is inferior to that of single task model even though the human parsing performance are improved. The reason may be due to very large imbalance with respect to loss levels. As shown in Fig. 2, the loss value of human parsing is nearly 1000 times larger than that of pose estimation. The present method sets loss weights by Eq. (10) based on the loss analysis of single-task models. The proposed loss weighting method considers not only the loss level but also the learning speed, balancing the training of both tasks to converge simultaneously. As shown in Table 8, the baseline model with only

Table 7
Pose estimation result comparison based on the validation set of the MPII dataset.

Method	Head	Sho.	Elb.	Wri.	Hip	Knee	Ank.	PCK↑
Newell et al. (2016)	96.5	96.0	90.3	85.4	88.8	85.0	81.9	89.2
Yang et al. (2017)	96.8	96.0	90.4	86.0	89.5	85.2	82.3	89.6
Tang et al. (2018)	95.6	95.9	90.7	86.5	89.9	86.6	82.5	89.8
SimpleBaseline (Xiao et al., 2018)	97.0	95.9	90.3	85.0	89.2	85.3	81.3	89.6
HRNet (Wang et al., 2020)	97.1	95.9	90.3	86.4	89.1	87.1	83.3	90.3
Hybrid-Pose (Kamel et al., 2020)	**97.5**	96.2	90.8	86.6	89.3	87.1	83.4	90.4
HTCorrM (Zhang et al., 2022c)	97.1	**96.4**	91.3	87.0	89.6	87.2	83.3	90.7
KDNet	97.2	96.3	**91.4**	**87.2**	**90.2**	**88.0**	83.5	**91.0**
KDNet w/o PE	97.3	96.2	91.0	87.0	89.9	87.3	**83.7**	90.8

Table 8
A comparison between single task model and MTL model with different loss weighting methods.

Type	Weighting methods	Mean IoU↑	PCK↑
STL	–	56.74	88.18
MTL	Equal weighting	56.21	84.41
MTL	Uncertainty (Kendall et al., 2018)	57.18	81.04
MTL	KD (Li and Bilen, 2020)	57.88	86.15
MTL	Baseline w/ weighting Eq. (10)	57.28	87.89
MTL	**KDNet** (full model w/ weighting Eq. (10))	**59.24**	**88.56**

Table 9
Ablation study on the effectiveness of cross-task representation consistency (CRC) module based on the LIP validation set.

	Mean IoU↑	PCK↑
Baseline	57.08	87.89
With CRC-HP	57.51	87.83
With CRC-PE	57.87	88.03

Table 10
Ablation study on the effectiveness of task-specific knowledge distillation (TRD and LD modules) based on the LIP validation set.

	Mean IoU↑	PCK↑
Baseline	57.08	87.89
With TRD	57.57	87.83
With LD	58.28	87.82
With TRD & LD	58.40	88.23
With TRD & LD & CRC-PE	59.24	88.56

the proposed loss weighting (without any of TRD, LD or CRC module) can achieve 87.89% PCK on pose estimation, outperforming equal loss weighting by 3.48% PCK, uncertainty-based weighting scheme (Kendall et al., 2018) by 6.85% PCK, the other KD-based scheme (Li and Bilen, 2020) by 1.74% PCK. Nevertheless, comparing to the single PE model, the performance of such baseline with proposed loss weighting is still 0.29% lower (87.89%–88.18%). There are two probable reasons for the inferior performance: firstly, the training of human parsing and pose estimation is not always balanced throughout the whole MTL training process; and secondly, there is no information exchange between two tasks and the relationship between them is not leveraged to assist each other. With strengthened task balance and information exchange between two tasks through knowledge distillation, the MTL model (KDNet) finally achieves 88.56% PCK on pose estimation and 59.24% Mean IoU on human parsing, outperforming single PE model by 0.38% and 2.5%, respectively.

It is interesting to find out whether the proposed loss weighting algorithm (Section 3.5) can be applied to and benefit other MTL methods fusing multiple tasks. HTCorrPM model (Zhang et al., 2022c), fused pose estimation with human parsing, was trained using the proposed loss weights obtained by Eq. (10). It is compared in Table 2 that HTCorrPM (Zhang et al., 2022c) with the proposed loss weights can lead to around 1% gain in performance than that without such weighting, improving the mean IoU from 56.85% to 57.86%. In the following sections, we will use the MTL model with weighting Eq. (10) as our baseline.

4.4.2. The effectiveness of the cross-task representation consistency module

The role of the CRC module is to allow the HP and PE representations to exchange information and enhance each other. “CRC-PE” represents the CRC module that transforms the pose representation to have similar pixel-wise relation with that of parse representation. Likewise, “CRC-HP” represents the CRC module that transforms the parse representation to have similar pixel-wise relation with that of pose representation. Experimental results show that by introducing CRC-PE module only, the performance of pose estimation is improved by 0.14% (88.03%–87.89%) PCK and the performance of human parsing is improved by 0.59% (57.87%–57.28%) Mean IoU over the baseline (see Table 9). Comparatively, the introduction of CRC-PE module results in larger performance improvement than the CRC-HP module does on both tasks, therefore, the CRC-PE module is chosen in the current KDNet structure.

4.4.3. The effectiveness of task-specific knowledge distillation

Experiments were conduct the measure the impact on the performance of human parsing and pose estimation of different knowledge distillation modules, namely, TRD and LD modules. As shown in Table 10, by transferring the representation of the single-task models to the KDNet using the TRD module, the mean IoU of human parsing has around 0.3% gain increases from 57.28% to 57.57%, while by distilling the prediction logits from the single-task models to the KDNet using the LD module, the mean IoU of human parsing is increased to 58.28% (0.7% gain), outperforming the baseline by 1.0%. The introduction of only TRD module (or only LD module) does not have a great influence on the pose estimation performance; the reason may be that the loss value of human parsing is much larger than that of pose estimation and it is difficult to change the biased situation by simply representation distillation or logit distillation. Nevertheless, introducing both modules, the mIoU of human parsing is improved to 58.40% and the PCK of pose estimation is improved to 88.23%, surpassing the single-task pose model. This illustrates the benefit of distilling knowledge of output logits in addition to intermediate features for both tasks of human parsing and pose estimation. After using the proposed CRC module, the mean IoU is increased to 59.24% and the PCK is increased to 88.56%, highlighting the efficacy of our full model.

In summary, TRD and LD modules distill knowledge from single-task models to the corresponding task branches of the KDNet, while CRC module strengths the combination between task-specific features, facilitating comprehensive information exchange.

4.5. Analysis and discussions

4.5.1. Computational complexity, efficiency versus accuracy

Table 11 compares the computational complexity, efficiency as well as the accuracy of KDNet and other state-of-the-art methods, following previous similar comparisons (Liu et al., 2022; Wang et al., 2020). Because the model size and FLOPs show minimal variation in cases of

Table 11
Comparison of model parameter numbers and computation costs based on the LIP dataset. All the experiments were conducted on one GPU of GeForce 3090. (The training and inference times for CNIF and HHP are unavailable because their codes cannot be executed on the 3090 GPU card, while the training time for JPPNet cannot be provided due to its training being immediate start and stop.)

Method	Backbone	Param↓	FLOPs↓	mIoU↑	PCK↑	Training time↓ (s)	Inference time↓ (s)	Task
CNIF (Wang et al., 2019)	ResNet-101	383.6M	300.0G	57.74	–	–	–	HP
HHP (Wang et al., 2021)	ResNet-101	258.6M	–	–	–	–	–	HP
CDGNet (Liu et al., 2022)	ResNet-101	71.1M	81.7G	59.04	–	0.493	0.016	HP
HRNet	HRNetV2-W48	68.7M	72.9G	56.74	–	0.235	0.033	HP
KDNet w/o PE encoder branch	HRNetV2-W48	68.7M	72.9G	59.24	–	0.507	0.033	HP
HRNet	HRNetV2-W48	68.7M	72.9G	–	88.2	0.256	0.033	PE
KDNet w/o HP encoder branch	HRNetV2-W48	68.7M	72.9G	–	88.6	0.507	0.033	PE
JPPNet (Liang et al., 2018)	ResNet-101	89.1M	263.5G	51.37	82.7	–	0.467	Both
MuLA (Nie et al., 2018a)	Hourglass	44.4M	43.4G	49.30	87.5	1.141	0.164	Both
NPPNet (Zeng et al., 2021)	ResNet-50	73.4M	113.7G	58.56	88.3	0.882	0.059	Both
KDNet	HRNetV2-W48	72.0M	101.1G	**59.24**	**88.6**	0.507	0.034	Both

Table 12
Performance comparison on different backbones based on the test set of the ePPP dataset.

Model–Backbone	Ref./KD	Params↓	FLOPs↓	Mean IoU↑	mAP↑
HP-W48	(a1)	68.65	72.89	70.01	–
HP-W32	(a2)	31.52	40.61	69.14	–
HP-W18	(a3)	10.81	19.70	68.20	–
PE-W48	(c1)	68.65	72.92	–	45.5
PE-W32	(c2)	31.53	40.64	–	46.4
PE-W18	(c3)	10.82	19.72	–	45.6
KDNet-W32	(a2)+(c2)	33.74	59.66	71.16	46.9
KDNet-W18	(a3)+(c3)	12.06	30.43	70.96	46.1
KDNet-W32	(a1)+(c1)	33.74	59.66	**71.37**	**47.2**
KDNet-W18	(a1)+(c1)	12.06	30.43	70.85	46.2

ePPP and MPII datasets, the comparison below is therefore based on the LIP dataset.

Since the parsing branch and pose branch are completely separate in KDNet, the pose branch can be discarded in the inference phase when deploying the model for parsing only, vice versa when deploying the model for pose estimation only. The present method only for one task has 68.7M model parameters, more than three times smaller than the size of hierarchical methods, like CNIF (Wang et al., 2019) and HHP (Wang et al., 2021). Compared to the most recent approach CDGNet (Liu et al., 2022), KDNet also has less model parameters and FLOPs yet with better parsing performance. In terms of training time, since the convolutional layers are executed in series on PyTorch and HRNet (i.e. our baseline) involves multiple branches operating at different resolutions simultaneously, the speed of HRNet is, therefore, slower than that of CDGNet, which is based on ResNet. Nevertheless, the training of KDNet can be further improved by implementing parallel convolutions. Furthermore, when focusing solely on human parsing or pose estimation, KDNet does not increase inference time or model parameters compared to HRNet, as the additional pose estimation (PE) encoder or human parsing (HP) encoder can be omitted.

Comparing the complexity of models for both tasks, KDNet has the shortest training and inference times among all the models. In addition, the present method outperforms JPPNet (Liang et al., 2018) by 7.87% mIoU and 5.9% PCK with less model parameters and FLOPs. Although MuLA (Nie et al., 2018a) has smaller model size and runs more efficiently, it is very much biased to the pose estimation task and only achieves 49.3% in terms of mIoU, representing a 9.94% drop than KDNet. Compared to HRNet (Wang et al., 2020) (the single-task model), KDNet for both tasks requires extra 3.3M (72.0M–68.7M) parameters and exceeds by 2.5% (59.24%–56.74%) mIoU and 0.4% (88.6%–88.3%) PCK. All these results demonstrate the efficiency of the proposed KDNet.

4.5.2. Wide applicability

The proposed KDNet has simple yet effective structure, can be applied to and integrated with other modules or encoders. Experiments were conducted on ePPP dataset and the results are given in Table 6. By adding an edge detection branch, it further improves about 0.3% of mean IoU compared to the basic KDNet. By replacing the HP encoder of KDNet with OCR module, it achieves 72.92% in terms of mean IoU, leading to 0.89% performance boost. When such model is trained using pose prediction results of Wu et al. (2019) rather than pose groundtruth, denoted as "KDNet–OCR w/o gt" in Table 6, it achieves almost the same performance as that of KDNet–OCR model. All these experiments have demonstrated the robustness of KDNet, which has highly adaptable structure and can work with other methods.

4.5.3. Different backbones

The proposed KDNet has a flexible structure that can be applied on different backbones. Experiments were conducted on ePPP dataset to train KDNet with smaller backbones of HRNetV2-W32 and HRNetV2-W18, respectively, and the results are shown in Table 12. As shown, KDNet with HRNetV2-W32 backbone increases the mean IoU by 2.02% than the single parse model of similar size (HP-W32) and increases the mAP by 0.5% than the single pose model of similar size (PE-W32). Based on the backbone HRNetV2-W18, KDNet improves the single parse model (HP-W18) by 2.76% mean IoU and the single pose model (PE-W18) by 0.5% mAP. Moreover, KDNet optimizes human parsing and pose estimation simultaneously without substantially increasing model size and computational costs. Specifically, the KDNet increases 2.21M parameters and 19.02G FLOPs on HRNetV2-W32, and increases 1.24M parameters and 14.71 FLOPs on HRNetV2-W18. In fact, the smaller size MTL models with knowledge distilled from STL models can outperform, in both tasks, the STL models. Moreover, compared to the KDNet by distilling knowledge from larger single task (STL) models of HP and PE trained on HRNetV2-W48 backbone, the KDNet by distilling knowledge from STL models trained on the same backbone almost achieves the same performance. This has further demonstrated that the proposed KDNet has the flexibility to apply on smaller models and still guarantees performance improvements of both tasks without increasing much parameters and FLOPs.

4.5.4. Parallel versus one-way distillation

Fig. 1 shows that KDNet is a MTL network with distilled knowledge from single task models of human parsing (HP) and pose estimation (PE). The experimental results have demonstrated that KDNet can simultaneously optimize both tasks. In traditional MTL tasks, it is more common to improve a target task by learning or sharing feature of an auxiliary task. It is therefore interesting to find out whether the proposed method is still effective for one-way knowledge distillation rather than parallel distillations. An experiment was conducted on MPII dataset for the pose estimation task. By distilling knowledge

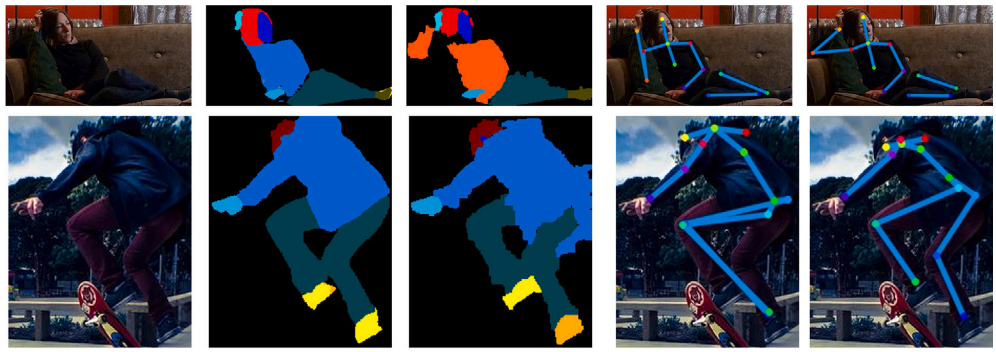

Fig. 5. Failure cases on the LIP dataset: (a) input images, (b) our parsing results, (c) parsing ground-truth, (d) our pose estimation results, (e) pose ground-truth.

only from a human parsing network (STL) trained offline on ePPP dataset, the performance of KDNet model (denoted as "KDNet w/o PE") is given in Table 7. It shows that such MTL model with one-way distillation still outperforms STL PE model (denoted as "HRNet Wang et al. (2020)" in the table) by 0.5% (90.8%–90.3%) of PCK even though it slightly underperforms than KDNet with parallel distillations. This again demonstrates that the proposed KDNet has a flexible structure and performance improvement can be obtained by distilling knowledge from either one task or both tasks.

4.6. Failure analysis and limitation

Fig. 5 shows some failure cases on human parsing and pose estimation. As shown, under severe lighting or occlusion conditions, the proposed KDNet fails to accurately identify the intended regions and joints, resulting in errors in both parsing and pose estimation. In addition, as our method adapted the offline knowledge distillation approach, the training time is about two times longer than the single model with the same backbone. As a future work, we will explore solving this problem, e.g. by using shadow knowledge distillation (Li and Jin, 2022). Moreover, graph neural network like Zhong et al. (2023) may also be utilized to model the relationship between human parsing labels and pose estimation joints.

5. Conclusion

Combined network training for human parsing and pose estimation is a non-trivial task. In order to achieve a good performance on both tasks simultaneously and efficiently in a multi-task learning (MTL) network, a knowledge distillation-based scheme is presented here to enhance the information exchange between human parsing and pose estimation while ensuring balanced training across both tasks. For information communication between specific tasks, the knowledge of the single-task models are distilled to the KDNet by the task-specific representation distillation and logit distillation modules. Because of the close relationship between human parsing and pose estimation, a consistent representation module between the representations of the two tasks is proposed. Furthermore, a loss weighting scheme is proposed to balance the loss levels based on single-task model trained offline. Detailed experiments have demonstrated that the proposed method can achieve remarkable performance on both tasks without sacrificing model complexity or computational efficiency during inference. As a future work, we will try to improve the discriminative power of representations and reduce the model training time, possibly through techniques such as self-supervision learning and shadow knowledge distillation (Li and Jin, 2022). Moreover, we will also explore generalizing the proposed method to a boarder range of tasks in the future.

CRediT authorship contribution statement

Yanghong Zhou: Writing – original draft, Methodology, Investigation, Conceptualization. **P.Y. Mok:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

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Data availability

I have shared the link to my data/code in the abstract.

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